

The Effects of Color-Coded Ambient Lighting on Task Switching, Workload, and Comfort

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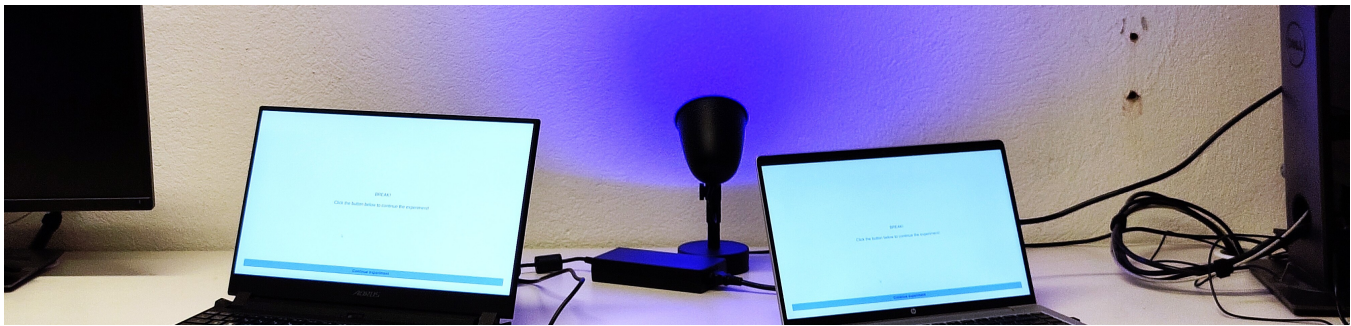


Figure 1: Controlled lab setup of Study I and Study II.

Abstract

Task switching is a common cognitive process that often disrupts focus and reduces productivity, particularly for people managing busy schedules or experiencing attention-related difficulties. This study investigates how colored ambient lighting can support smoother transitions between tasks by providing subtle visual cues for different workflow scenarios. We designed an experimental setup where participants performed one primary task alongside two periodically changing secondary tasks, under two conditions: one with colored, dynamic ambient lighting and one with fixed ambient lighting. However, our findings suggest that there is no evidence for smoother task transitions or increased well-being scores reported by participants. By comparing user performance and subjective comfort across conditions, we aim to evaluate whether ambient light cues can enhance well-being, comfort, and overall user experience during task transitions.

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CCS Concepts

• **Human-centered computing** → *Accessibility systems and tools; Ambient intelligence; Empirical studies in interaction design.*

Keywords

Task Switching, Context Switching, Light-aided, Concentration.

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1 Introduction

Task switching is a pervasive feature of digital work, yet even predictable shifts between tasks carry measurable “switching costs” in speed, accuracy, and mental effort [2, 18, 20, 22]. These costs can diminish with practice as people learn task sequences and cues, but repeated switching still tends to disrupt focus, especially when the environment offers little support for anticipating upcoming changes [31, 33, 34]. At the same time, research on ambient illumination shows that light characteristics can modulate alertness and cognitive performance, suggesting that lighting might be used not only for comfort but also as a subtle layer of cognitive support [4, 9, 17, 19, 32].

This work investigates whether color-coded ambient lighting can shape how people adapt to repeated task switching over time.

Grounded in existing literature, we proposed the hypothesis for this paper as follows:

Participants will report a higher level of comfort under automated, task-adopted, colored environment lighting compared to uniform, static, white environment lighting.

The formulation of above hypothesis is grounded in previous work from Hartstein et al., which shows that changing color temperature of lighting leads to a significant (15.2%) improvement in task switching accuracy for preschool children [12]. Furthermore, Xu et al. also suggest that people can associate environmental cues with the need for cognitive flexibility [35], providing a fundamental argument for the stated hypothesis. The hypothesis is further reinforced by other authors who show the impact of environmental lighting on intrinsic human sensations such as well-being, cognition, mood, and alertness ([4, 17, 19, 32]).

The hypothesis is also grounded in Aliakseyeu et al.'s paper on intelligent environment changes [1], which by exploring different possibilities of automated adoptions of lighting environments provide technological rooting for our work. This is further consolidated by Hwang et al. who propose a lightweight LED smart switch, helping to pave the way for any product development originating from this paper [15].

Additionally, following research questions were stated to be answered in this paper:

- Q1** How can cognitive task switching be supported by providing environments differentiated by color?
- Q2** Do differently colored environments lead to a more enjoyable work setting in multitasking scenarios?
- Q3** How does the comfortability of users change with introducing color-based work environments?

In a controlled within-subject lab study, participants ($n=20$) each completed two runs of a multitasking scenario, once under fixed white ambient lighting and once with color-coded lighting that differentiated the side tasks, with the order counterbalanced so that half experienced the colored-light condition first and half second. By subjective experience across runs and conditions, the study focuses on whether improvement over the two tests is more pronounced when color cues are present, and whether any such gains can be attributed to the lighting manipulation rather than mere practice.

We used neutral white light as the baseline condition, as white lighting is the dominant standard in office and study environments and is frequently rated as most suitable for productive cognitive work [37]. For the color-coded condition, blue and green hues as representatives of cool workplace colors were selected. We further backed the choice of color in the precedent work of Chong et al. who gave evidence for color-specific cognitive effects. In their paper they show that blue environments show the highest memory task accuracy and green environments produce the fastest reaction times for attention tasks, which is why we select this color combination. They further argue that yellow light leads to faster working-memory performance and response times [5]. However, we decided against using yellow light due to its similarity to the neutral white light appearing in naturally warmer hues. The choice

of using blue and green was also beneficial due to considerations regarding colorblindness.

Prior work on color in work environments associates blue and green with calmness, attentional focus and a sense of well-being, in contrast to more arousing warm colors. In addition, studies on blue-enriched lighting report improvements in alertness and reaction times in cognitively demanding tasks [28, 37]. On this basis, blue and green lighting were theoretically well-motivated candidates for supporting focused task switching, while white light served as a familiar control condition.

The contribution of this work is to combine a rigorous experimental design with an ecologically inspired task setup to examine how a simple, deployable environmental change, ambient color cues, can influence learning, efficiency, and perceived effort in task switching. The following sections describe the methods in detail, present the quantitative results of the two-run comparison, and discuss what these findings imply for designing lighting systems that do more than illuminate: they also help the mind keep up with a demanding flow of tasks.

The chosen ambient light color was selected to act as a subtle, office-plausible cue for task context rather than a strong mood or arousal manipulation, aligning with work showing that moderate variations in lighting color temperature can influence alertness, workload, and perception without producing large or robust performance effects, and that highly saturated or extreme colors risk confounding results through emotional and comfort changes rather than task-related mechanisms.

2 Method

This work reports on two complementary studies: a quantitative experiment (Study I) as shown in Figure 2 and a qualitative follow-up (Study II) that together examine how color-coded lighting affects task switching, workload, comfort, and flow in a multitasking workstation scenario.

2.1 Participants

A total of 20 participants took part in the studies (10 female, 10 male), with ages ranging from 20 to 34 years old ($M = 24.45$, $SD = 3.41$). All 20 participants completed Study I, with 19 of the participants additionally taking part in the follow-up qualitative interviews in Study II.

Participants were recruited over a period of approximately one to two weeks through in-class announcements and an online sign-up document provided as part of the course.

Inclusion criteria required basic computer skills, normal or corrected-to-normal vision, and sufficient English proficiency so that individual differences in fundamental abilities do not overshadow the effects of lighting. Exclusion criteria included self-reported color vision deficiencies and photosensitive epilepsy to ensure participant safety and that the lighting manipulation was perceptible.

All participants provided informed consent prior to participation. They were presented with an informed consent form produced by the Frankfurt University of Applied Sciences [23], which outlined the study's objectives, procedures, and their rights as participants. Before beginning the experiment, data from all participants were included in the analysis, as there were no technical issues that

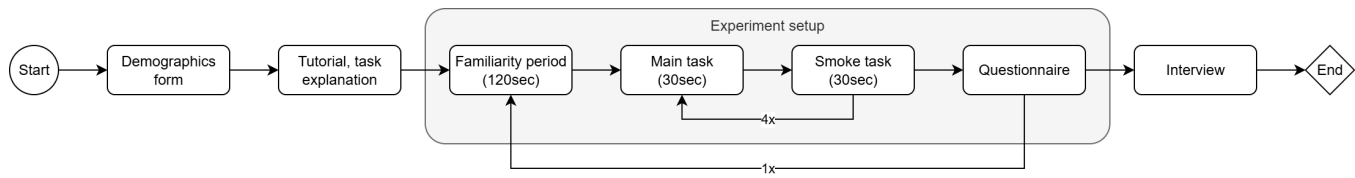


Figure 2: Schematic overview of the quantitative experiment.

required exclusion. Participants were compensated with a small treat.

2.2 Study Setup

The following section describes the shared experimental setup, tasks, and procedure used across both Study I and Study II.

The apparatus recreates an office-like setting using a laptop, a desk and chair, and an LED light source capable of displaying white, blue, and green light. Participants interact with a typing “primary” task (see Figure 4a) and two shorter cognitive “secondary” tasks (see Figure 4b) under carefully controlled lighting states, and all task performance and questionnaire data are collected using specially developed software. The design of the “primary” task is inspired by Caitlin McMullin’s 2021 paper “Transcription and Qualitative Methods: Implications for Third Sector Research” [21]. “Secondary” tasks 1 is implemented as a *Spot-the-Difference game*, similar to Fukuba et al. [8] and Gross et al. [10]. “Secondary” tasks 2 is a *odd/even and vowel/consonant detection task*, which is a common approach throughout literature [24, 25, 30].

In an effort to allow for exact replication of the setup, the software is provided on Github¹. Additionally, the exact questionnaires (incl. resulting data) are provided in Appendix A.

The procedure follows a fixed sequence: informed consent and demographics, task explanation and familiarization, followed by one 8-minute block of interleaved primary and secondary tasks under one lighting condition. The alternation of tasks was timed to be 30 seconds, which accounts for the required cool-down period of micro-breaks (the “primary” task in this case), as mentioned by Hennfng et al. [14]. The procedure then continues with questionnaires, a break, then the same block under the other lighting condition, followed by a brief interview. This step-by-step protocol, applied identically to all participants with documented deviations, supports procedural rigour and reproducibility.

2.3 Study I: Quantitative Evaluation of Color-Coded Lighting for Task Switching

Study I is a within-subject quantitative evaluation that tests whether assigning distinct background colors (see Figure 3b) to different tasks can reduce perceived workload, increase comfort and flow, and improve task performance compared to a uniform white-light (see Figure 3a) baseline. A controlled lab study is appropriate here because it allows precise manipulation of lighting conditions and task sequences, systematic control of confounds (e.g., order effects via counterbalancing), and the use of established instruments to capture both workload and experiential quality. Specifically, the

NASA Task Load Index (NASA-TLX) [11] and the Flow Short Scale [27] were employed to measure subjective mental workload and flow, respectively. With flow being seen as an indicator of intrinsically rewarding and psychologically positive task engagement that is closely related to subjective comfort and well-being during activity [7, 13].

Alternative designs such as purely between-subjects comparisons or uncontrolled field deployments were dismissed at this stage because they would either require substantially larger samples or provide less experimental control for a first, proof-of-concept evaluation, and because differences in participants’ task-related skills could otherwise confound the effects of the lighting conditions in a between-subjects design.

2.4 Study II: Qualitative Exploration of Lighting, Flow, and Comfort

Study II is a qualitative follow-up that uses semi-structured interviews to explore how participants experienced the different lighting conditions, moments of comfort or discomfort, and states of flow and distraction during the task-switching scenario. A qualitative design is appropriate here because it can address open questions left by Study I, reveal mechanisms behind the quantitative patterns, and surface unanticipated themes (e.g., feelings of support, aesthetic reactions, or subtle distractions) that are not captured by scales and performance metrics alone.

The apparatus in Study II largely mirrors that of Study I, allowing direct comparison and integration of findings; in the report, overlapping aspects such as the lab setting and hardware can be explicitly referenced as “as in Study I” to avoid redundancy while preserving reproducibility.

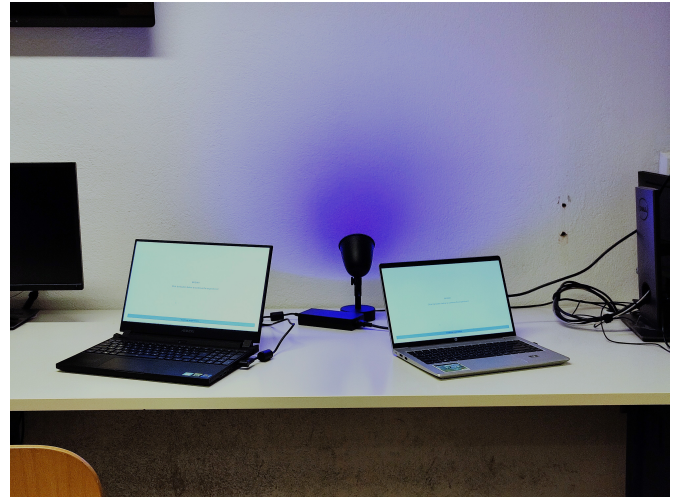
Oriented by an established five-phase framework for interview guide development [16], the study subsequently employed a thematic analysis, the team develops, pilots, and refines an interview guide with open, non-leading questions (provided in Appendix B), then records, transcribes, and systematically codes the data to derive themes that speak back to the original research questions.

By structuring the project as a quantitative experiment followed by a qualitative deep dive, the work is anchored in a mixed-methods logic that uses each method for what it does best: estimating the size and reliability of effects (Study I) and understanding participants’ perspectives and sense-making (Study II). Together, these studies provide a more complete, methodologically grounded account of how color-coded lighting can shape task switching, workload, comfort, and flow in everyday computer-based multitasking.

¹<https://github.com/johannesgaebler/tuw-urm-study>, accessed on 14.01.2026



(a) The experiment setup during the primary task (white light).



(b) The experiment setup during the secondary tasks (blue light).

Figure 3: The experiment setup.

Qualitative data were analyzed using an inductive thematic analysis approach [3]. The transcripts² were openly coded using the Taguette tool³, allowing themes to emerge from the data rather than being predefined. After initial coding, the coders discussed and reviewed the codes to ensure internal consistency and refine emerging themes. The final themes were developed through iterative comparison and consolidation of the coded data.

3 Results: Study I - Quantitative

The study followed a mixed-methods design. This section reports the quantitative results, focusing on participants' subjective task experience under two lighting conditions (colored light vs. white light). Outcomes were assessed using perceived workload (NASA-TLX) [11] and components of the Flow Short Scale [27].

All statistical analyses were conducted in R (RStudio). The significance level was set at $\alpha = 0.05$. Data completeness was achieved across the sample of 20 participants across both lighting conditions, and no missing values were present in the questionnaire measures. Each participant contributed data to both conditions, resulting in a fully paired dataset.

3.1 Perceived Workload (NASA-TLX)

Perceived workload was assessed using the NASA-TLX [11]. For each participant and lighting condition, an overall workload score was computed by averaging the six sub-scale ratings. Differences between lighting conditions were analyzed using within-participant comparisons.

The distribution of within-participant difference scores (colored minus white lighting) was examined prior to hypothesis testing. Visual inspection of histograms and normal Q-Q plots suggested no substantial deviations from normality, which was supported by

a Shapiro-Wilk test, $W = 0.91$, $p = 0.074$. Accordingly, differences between lighting conditions were analyzed using a paired-samples t-test.

Mean workload was 9.75 in the white-light condition and 9.85 in the colored-light condition. The paired-samples t-test showed no statistically significant difference between conditions, $t(19) = 0.25$, $p = 0.81$. The mean difference was small ($M = 0.10$), with a 95% confidence interval ranging from -0.74 to 0.94. Although individual workload ratings varied across participants, no systematic trend favoring either lighting condition could be found. Overall, these results suggest that colored lighting did not influence participants' perceived workload during the task.

3.2 Flow Short Scale

The Flow Short Scale [27] was used to assess flow experience, task-related worry, and perceived skill-demand balance. In line with the scales' validated structure, these components were analyzed separately. Results for each component are reported below.

3.2.1 Flow Experience. Flow experience was assessed using a combined score computed as the mean of Items 1-10 of the Flow Short Scale. Higher scores indicate a stronger subjective experience of flow during task performance. Mean flow scores were similar across conditions, with slightly lower values under colored lighting ($M = 4.76$) compared to white lighting ($M = 4.84$).

Inspection of within-participant difference scores indicated deviations from normality, as reflected in histogram and Q-Q plot inspection and confirmed by a Shapiro-Wilk test, $W = 0.89$, $p = 0.024$. Accordingly, differences between lighting conditions were analyzed using a Wilcoxon signed-rank test. The analysis revealed no statistically significant difference in flow experience between the colored-light and white-light conditions, $V = 70$, $p = 0.32$. The median difference score was -0.25, and the effect size was small ($r = 0.23$). As illustrated in Figure 5, individual participants showed

²Full transcripts are available at <https://github.com/johannesgaebler/tuw-urm-report/blob/main/Data/InterviewTranscripts.txt>, accessed on 14.01.2026

³<https://www.taguette.org/>, accessed on 14.01.2026

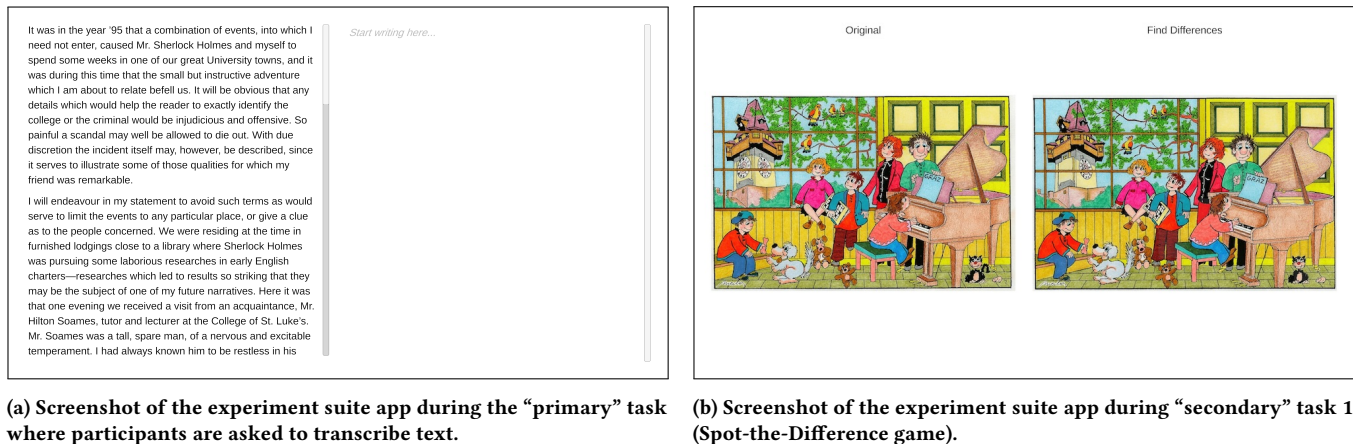


Figure 4: Screenshots of the experiment suite app.

varied changes in flow across lighting conditions. While many participants exhibited small decreases in flow under colored lighting, a smaller number showed clear increases, resulting in no consistent group-level pattern.

Overall, these results indicate that colored lighting did not reliably influence participants subjective experience of flow during the task.

3.2.2 Task-Related Worry. Task-related worry was assessed using the Worry sub-scale of the Flow Short Scale (Items 11-13). A composite worry score was computed for each participant and lighting condition by averaging the three items. Mean worry scores were low in both conditions, with slightly lower values under colored lighting ($M = 2.35$) compared to white lighting ($M = 2.48$).

The distribution of within-participant difference scores (colored minus white lighting) was examined using histograms and normal Q-Q plots and did not show substantial deviations from normality, which was supported by a Shapiro-Wilk test ($W = 0.91$, $p = 0.064$). Differences between lighting conditions were therefore analyzed using a paired-samples t-test.

The paired-samples t-test revealed no statistically significant difference in task-related worry between the colored-light and white-light conditions, $t(19) = -1.36$, $p = 0.19$ (95% CI $[-0.34, 0.07]$), indicating no meaningful effect of lighting condition on task-related worry.

3.2.3 Skill-Demand Balance. Perceived balance between task demands and participant skills was assessed using the Skill-Demand Balance items of the Flow Short Scale (Items 14-16). These items were analyzed descriptively. Mean skill-demand balance scores were similar across lighting conditions, with values close to the midpoint of the scale in both the white-light ($M = 4.87$) and colored-light ($M = 4.62$) conditions. This suggests that the task was perceived as neither overly easy nor overly demanding under either lighting condition.

4 Study II: Qualitative Results

The study followed a mixed-methods design and the sample size was $n=19$. This section reports the qualitative interview data which was

analyzed separately to capture participants’ experiential accounts. Inductive thematic analysis yielded 21 codes, from which three overarching themes were derived.

4.1 Theme 1: Lights influence in task and emotion

The analysis showed that the participants exhibited mixed responses to lighting conditions. Where Some participants remained unaware of lighting during task performance, others reported either distraction or positive engagement with the lighting environment. one participant said,

I didn’t notice the color changing, but that went by really fast. It was like, okay, there’s something going on in the background, and then I was back, you know, concentrated into the... Overall, I have to say, I didn’t notice them too much. I noticed the lights changing, but nothing about the color or temperature. (P5)

This participant’s sustained task focus despite peripheral awareness of lighting changes suggests that chromatic variations did not substantially disrupt attentional engagement or task-related cognition. In contrast, another participant reported a notably different experience:

It helps. It reduced the stress because I knew what was coming before it came. So during the switch, I was less stressed. Yeah. (P8)

This participant noticed the color changes and found them beneficial, reporting that chromatic shifts served as anticipatory cues that reduced stress during task transitions. Chromatic shifts may facilitate cognitive preparation and reduce transition-related stress for some users. While chromatic shifts may facilitate task transitions for some participants, others experienced them as disruptive. One participant reported:

Yes, I think with the changing lights, they were distracting sometimes, because it just drew my attention over to the colored light instead of to what I was doing, and then often it was timed with the change of the task, so

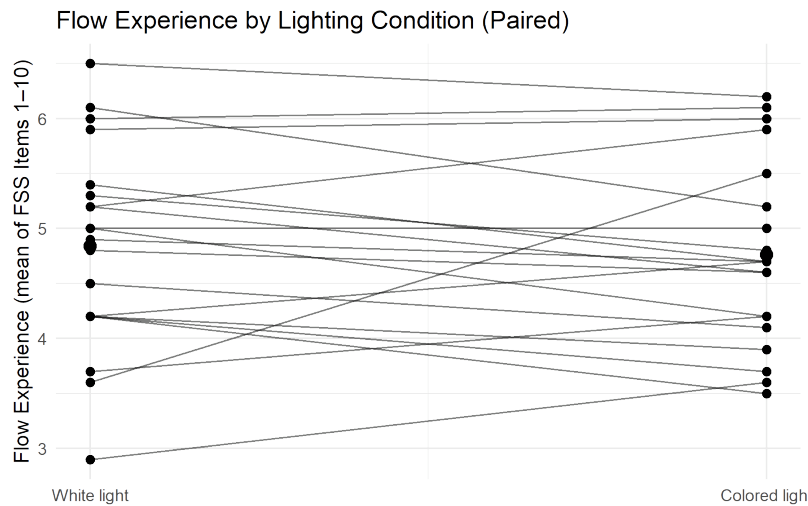


Figure 5: Flow experience scores under white-light and colored-light conditions. Each line represents one participant, connecting their flow score under white light (left) and colored light (right).

then I was, I think, lost for a bit longer, and it just took a bit longer to readjust to what was going on. (P17)

For this participant, chromatic changes induced attentional capture that disrupted task focus, particularly during task transitions. The timing of lighting changes coinciding with task switches compounded this effect, prolonging the reorientation period and hindering the maintenance of cognitive flow. In response to preferences between a chromatic shift environment and a single-color environment, one participant stated:

I think that changing the light distracts me more. When the light was the same during the whole time, I was more concentrated on the tasks that I needed to do, and it was kind of a surprise for me to see a new color. (P2)

This participant preferred a single-color lighting environment for maintaining concentration. In contrast, another participant expressed a positive experience with chromatic shifts, stating:

But when it changed with the task, it felt nice. It felt like going to a different environment for a different task, even though I sat in the same spot. (P14)

Overall, the results show that participants reacted differently to the lighting conditions. Some participants hardly noticed the lighting changes and were able to stay focused on their tasks, while others were clearly affected by them. For some, chromatic shifts were helpful, as they reduced stress, supported task transitions, and created a sense of moving into a new context. For others, the same lighting changes were distracting and made it harder to stay concentrated or quickly adjust to a new task. These mixed responses suggest that chromatic lighting does not work in the same way for everyone.

4.2 Theme 2: Color and lighting preferences

Participants' acceptance of color-coded ambient lighting depended less on the idea of cueing and more on subjective comfort and preference such as hue associations, brightness tolerance, retinal

fatigue, varying thresholds for distraction and whether dynamic changes felt appropriate in a “work-like” setting. These preferences shaped whether lighting was experienced as supportive, neutral or distracting.

Participants described colors in emotional terms (relaxing, stressful, official), showing that color operates as a psychological signal, not a neutral variable.

Blue, the red color makes me more stressed. (P3)

Green was very distracting for me; I don't know why. (P8)

Then it was more comfortable, because I felt more confident in what I was doing. And then I think a couple times when it was switching between tasks, and I had the blue light, it was a bit less comfortable, and it was throwing me off my tracks. (P17)

Several answers from the participants show that brightness and saturation can dominate impressions as well. One participant constantly rubbed his eyes after the experiment and noted that bright light could have contributed to the discomfort in his eyes.

I *How did the color changes affect your focus?*

P2 *And I also feel like when it switched to blue somehow, I was... A little uncomfortable.*

I *Are your eyes hurting right now?*

P2 *Yeah, a little bit. I felt them a little.*

P8 also was distracted by the light:

I found the light very disturbing in some way. Because I was like, when I looked at it, it was too bright for my comfort. (P8)

On the other hand, one participant noted that the blue color was an additional factor for concentration. However, such feedback about the positive effect of blue on task performance was significantly less common than negative and distracting descriptions.

I like blue one, because white one was more official. Actually, I would say it was more, not stressful, but rather focused. Stressful is not the right word, but it gives the understanding that it was more locked in.
(P14)

Regarding the lighting condition preferences, stable lighting was perceived as more “work-appropriate” for some users. A recurring pattern is that some participants questioned whether dynamic ambient lighting belongs in a realistic work environment and preferred one stable condition.

If you're asking if I would rather work in a room where the light is changing versus a room where the light isn't changing, give me a room where the light isn't changing because why would the light change in the workplace?
(P13)

Moreover, one participant described warm or softer light as more comfortable and contrasted both of them directly:

Because I think the second light was warmer, and the first one was colder, and I feel like with the warmer it's better for me. (P12)

At the same time, a subset of participants explicitly preferred the colored setup, describing it as more pleasant or engaging than plain white. One participant, who normally uses ambient orange and red lighting at home, stated that they would prefer to do similar tasks “with color” rather than in a standard, white-lit room.

Maybe because, for example, at my place I don't use the main light which is normally white or yellowish a bit. I use ambient lighting, and I use also orange a bit a bit of yellow or red and I think I work better having this light. (P7)

4.2.1 Subtheme: Negative color afterimages. One of the most practically relevant and unexpected findings is that participants sometimes struggled to recall which colors appeared or reported colors that were not used, suggesting the cue mapping may not have become stable knowledge for everyone. Several participants mentioned the color that was not used in the setting:

Was there a purple one? I don't remember. Was it like pink? I remember it's so purple. And I was like, oh, that's really beautiful. Then it was gone. (P8)

Yeah. There was pink, yellowish, and just bright. ... Or maybe blue or magenta. (P1)

This pattern is consistent with classic findings on negative color afterimages. When observers fixate a colored area for some time and then look at a neutral field, in our case it was white light, they often perceive a lingering patch in the complementary color of the inducer [36]. These phenomena are commonly explained within opponent-color theory, which models color perception as the output of red-green and blue-yellow opponent channels in the visual system [26, 29]. The resulting imbalance is perceived as the complementary hue (magenta/pink), which matches participants' descriptions of “purple” or “pink” even though no such light was presented.

Overall, the theme shows that participants did not respond to “colored light” with a universal and predicted reaction, even when

the mapping is consistent, different affective interpretations can create opposite experiences for the same manipulation. Their comfort and preferences depended on hue, warmth, intensity and how closely the lighting matched their own habitual work environments.

4.3 Theme 3: Task performing experience

Participants shared a range of experiences related to the tasks, the study environment, and the overall study procedure. When asked which task they liked the most, one participant described the spot-the-difference task as particularly enjoyable and engaging:

Apart from that, I don't know, I always liked doing the difference spotting, I can't use that for anything like that, but no, I like that task. (P6)

While this task was perceived positively, participants also reflected on challenges related to moving between different task types. In particular, one participant noted that switching from an image-based task back to a text-based typing task was difficult:

It's a good question. It was harder for me to switch from the picture back to the text. And I think everything else was okay. But I noticed it was harder to switch back from the picture because I couldn't find myself in the text. (P8)

This response suggests that returning to typing required additional cognitive effort, as participants needed to reorient themselves and locate their previous position in the text. Based on this observation, future studies may benefit from selecting tasks that involve less cognitive load during transitions.

Beyond task design, participants also commented on how the study setting influenced their behavior. One participant explained that their performance changed depending on the pace of nearby participants:

I mean, same tasks, but behave differently. And mainly not due to the experience, but due to the people which I sat near. So, on the first task it was like I saw the guy typing very fast, I tried to type faster or same pace. On the second time I see that he's not so fast, so it was more chill. (P14)

This highlights the role of external social influences in shaping task performance and stress levels. Although individual study sessions could reduce such effects, practical constraints related to time and participant availability required the study to be conducted in a shared environment.

In addition to social factors, participants also pointed out technical aspects that affected their performance. One participant mentioned the need to adjust to an unfamiliar keyboard:

Yeah, at first I thought... At first I have to, like, always kind of get used to the keyboard of a new laptop. I think that's because I was somehow faster in the second iteration. (P2)

This indicates that adapting to a new keyboard, particularly for participants accustomed to a different layout (such as English versus German), can influence task speed and accuracy. We further discuss this limitation in subsection 5.5.

Overall, these reflections underline how both task design and environmental factors shaped participants' experiences

5 Discussion

5.1 Implications of Quantitative Findings

The quantitative results indicate that, across the measured subjective outcomes, no reliable differences emerged between the colored-lighting and white-lighting conditions. Neither perceived workload (NASA-TLX) nor components of the Flow Short Scale showed statistically significant changes as a function of lighting condition. This suggests that, within the context of the present task-switching scenario and exposure duration, colored ambient lighting did not substantially alter participants' subjective experience as a collective.

While the sample size of 20 participants limits the statistical power to detect very small effects at the group level [6], inspection of individual-level changes suggests a more varied pattern. As illustrated in the paired flow scores (see Figure 5), some participants reported increases in flow under colored lighting, whereas others showed small decreases, resulting in no consistent group-level trend. This pattern indicates that the impact of colored lighting may be highly individual rather than uniform, with potential benefits for some participants offset by neutral or negative effects for others. Therefore, any overall effects of colored lighting on flow are likely modest and strongly controlled by individual differences.

5.2 Implications of Qualitative Findings and Influencing Factors

End-of-session interviews revealed that several participants misinterpreted the color meanings or did not consistently notice the mapping between tasks and light, and in some cases even hinted at difficulties distinguishing colors which directly affects how strongly the lighting manipulation could influence behavior. Furthermore, this raises the possibility of unreported color-vision issues or color-perception changes over time when prolonged exposure to saturated colored light can induce negative afterimages, so that when the illumination returns to neutral white participants may briefly experience complementary hues (in our case described as “pink” or “purple” and mentioned in subsubsection 4.2.1) rather than our intended white baseline [26]. Recent work in vision science shows that such chromatic adaptation and afterimages can systematically shift perceived colors over seconds to minutes, even when the physical illumination is constant, which limits how reliably color alone can function as a stable task cue [36].

5.3 Potential ethics issues

Using ambient colors to steer attention and task transitions involves subtle influence, which becomes ethically sensitive when users cannot reliably perceive or interpret the cues. The interviews suggest that some participants misunderstood the role of color or may have had undiagnosed color-vision limitations, meaning that a system based on implicit color signals could inadvertently exclude or confuse certain users, especially if it is framed as universally supportive. To mitigate such risks, future designs should include explicit onboarding that explains the mapping, offer alternative or redundant cues (e.g., brightness, position, or gentle sound), allow users to self-declare color-vision issues and adapt the palette accordingly, and make sure that ambient cues remain optional aids

rather than hidden channels for performance nudging or opaque monitoring.

5.4 Social impact

If color-coded ambient lighting for task switching were widely adopted, it could become part of how people structure focus and transitions in shared and personal workspaces, potentially helping many users to mentally “gear up” for switches while keeping the environment calm and unobtrusive. However, the interview findings underscore that not everyone perceives or interprets colors in the same way, so a one-size-fits-all color system might reinforce disparities — for instance, people with color-vision deficiencies or different cultural associations might not benefit, or could even feel disadvantaged if expectations about responsiveness and multitasking are implicitly tied to reacting to light cues. Any large-scale deployment should therefore treat ambient lighting as an assistive option that can be personalized and negotiated, rather than as a universal layer of invisible control over how people work.

5.5 Limitations

Several limitations constrain how far these findings can be generalized. The sample size was relatively small ($n=20$ for Study I and $n=19$ for Study II), which limits statistical power to detect very small effects. In addition, the quantitative study was explicitly designed to compare lighting conditions rather than to examine learning or order effects across repeated task runs. As a result, only two task blocks per participant were collected, which does not provide grounds for a more detailed analysis of how performance or experience might evolve over time or differ depending on whether participants started with or without colored lighting. While such dynamics may be relevant for understanding longer-term adaptation, they were outside the scope of the present study. Moreover, the discovery in interviews that some participants misread or barely noticed the colors suggests that the effective strength of the independent variable (recognizable color-coded cues) was likely lower and more heterogeneous than intended, weakening internal validity and making it harder to detect clear effects.

5.5.1 Technical Limitations. During the execution of the experiment, various technical issues emerged which pose limitations to the results presented in this paper. The problems faced were mostly related to connectivity issues with the used light-apparatus, which created partly uncontrolled experiment conditions for some participants. Another technical limitation is that participants were using different devices to conduct the study on. In total, we had 4 different devices which all differed in size, keyboard layout and even operating system.

5.5.2 Conducting Limitations. Besides the technical limitations mentioned above, there were some complications related to the execution of the experiment. One of these was the uneven distribution of participants due to timing problems, which led to P10, P11, and P12 conducting the experiment at the same time.

5.6 Future work

Future studies should explicitly stratify participants by initial condition (starting with or without colored light), collect more than two

runs per person, and use cleaner logging so that learning curves and early exposure effects can be modeled with greater precision. Building on the interview insights, it will also be important to design and test variants that adapt to users' color perception — e.g., configurable palettes, alternative cue dimensions, or simple calibration tasks — while combining quantitative measures with richer qualitative accounts to better understand how people actually notice, interpret, or ignore ambient cues over time. Finally, extending this work beyond a single typing task to more varied real-world activities and incorporating explicit attention profiles (including confirmed color-vision status and neuro-diversity) would help clarify for whom, and under what conditions, ambient lighting can genuinely support smoother task switching without creating new accessibility or ethical concerns.

6 Conclusion

This work examined whether color-coded ambient lighting can ease cognitive task switching. The results show that its effects are subtle, inconsistent, and not robust enough to claim general benefits over standard white light. Across the quantitative measures, colored light neither reliably reduced perceived workload nor consistently enhanced flow or task-related worry compared to a white-light baseline. No robust improvements attributable to the lighting manipulation were observed across quantitative measures.

Qualitative interviews revealed that color cues were experienced along a spectrum from supportive, acting as anticipatory signals that reduced stress during switches, to distracting, drawing attention away from the task, with strong influences of individual color preferences, brightness tolerance, and even misperception such as confusion about which colors were present. Together, these findings suggest that ambient color cues can support smoother task transitions for some users under some conditions, but they do not operate as a universally helpful aid and may introduce new accessibility and comfort issues if deployed without personalization. Future lighting-based support systems for multitasking should therefore treat color as an optional, configurable layer of assistance, complemented by alternative cue dimensions and explicit on-boarding, and should be evaluated with larger, more diverse samples and longer exposure to clarify for whom and under what circumstances such cues genuinely improve task switching comfort.

The key takeaway the reader can remember is that ambient color is not a general-purpose comfort, nor productivity booster: its benefits for task switching depend strongly on individual perception and context, so only personalized and carefully designed color cues have the potential to improve comfort without backfiring.

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A Questionnaires

A.1 Questionnaire Questions

A.1.1 Part 1: Demographics.

- (1) Age
- (2) Gender
- (3) Profession
- (4) Highest education level
- (5) Do you have a history of photosensitive epilepsy or seizures triggered by visual stimuli (e.g., flashing or changing lights)?
- (6) Do you have a diagnosed colour vision deficiency (e.g., red-green colour blindness)?
- (7) How is your vision during computer work (with glasses or contact lenses if you normally use them)?

A.1.2 Part 2: NASA Task Load Index.

- (1) How mentally demanding was the task? Please select a number between 1 and 20. 1 = low, 20 = high
- (2) How physically demanding was the task? Please select a number between 1 and 20. 1 = low, 20 = high
- (3) How hurried or rushed was the pace of the task? Please select a number between 1 and 20. 1 = low, 20 = high
- (4) How successful were you in accomplishing what you were asked to do? Please select a number between 1 and 20. 1 = very unsuccessful, 20 = very successful
- (5) How hard did you have to work to accomplish your level of performance? Please select a number between 1 and 20. 1 = not hard at all, 20 = very hard
- (6) How insecure, discouraged, irritated, or stressed did you feel during the task? Please select a number between 1 and 20. 1 = not at all, 20 = very

A.1.3 Part 3: Flow Short Scale. The following questions were presented on a 7-point Likert scale.

- (1) I feel just the right amount of challenge. (1 = strongly disagree, 7 = strongly agree)
- (2) My thoughts/activities run fluidly and smoothly. (strongly disagree, strongly agree)
- (3) I don't notice time passing. (strongly disagree, strongly agree)
- (4) I have no difficulty concentrating. (strongly disagree, strongly agree)
- (5) My mind is completely clear. (strongly disagree, strongly agree)
- (6) I am totally absorbed in what I am doing. (strongly disagree, strongly agree)
- (7) The right thoughts/movements occur of their own accord. (strongly disagree, strongly agree)
- (8) I know what I have to do each step of the way. (strongly disagree, strongly agree)
- (9) I feel that I have everything under control. (strongly disagree, strongly agree)
- (10) I am completely lost in thought. (strongly disagree, strongly agree)
- (11) Something important to me is at stake here. (strongly disagree, strongly agree)
- (12) I won't make any mistake here. (strongly disagree, strongly agree)
- (13) I am worried about failing. (strongly disagree, strongly agree)
- (14) Compared to all other activities which I partake in, this one is ... (easy, difficult)
- (15) I think that my competence in this area is ... (low, high)
- (16) For me personally, the current demands are ... (too little, too high)

A.2 Result Data

We made all data available as part of the report repository on Github⁴.

B Interviews

B.1 Semi-Structured Interview Guideline

B.1.1 Intro questions.

- How did you experience the tasks overall today?

B.1.2 Light.

- What did you notice about the lighting during the tasks in each condition?
- Do you remember the different light colors?
- How did you feel under coloured light?
- How did you feel under white light?

B.1.3 Comfort and well-being.

- Can you describe any moment when the environment felt especially comfortable or uncomfortable?
- How did you feel?
- In what context?
- How did the light setting contribute to that?

⁴<https://github.com/johannesgaebler/tuw-urm-report/blob/main/Data>, accessed on 14.01.2026

- Would you prefer one lighting condition over the other for doing this kind of task? Please explain.

B.1.4 Flow.

- (If not mentioned already) If you compare the two lighting conditions, in which did you feel more ‘in the zone’ and why?
- Were there moments when you felt ‘lost’ or had to pause to reorient yourself? What triggered that?

B.1.5 Follow-up questions.

- Can you tell me more about that moment?
- What exactly made you feel that way?
- How did that compare to the other condition?
- Was there anything unexpected about your feeling or performance?

B.1.6 Closing question.

- Is there anything else you’d like to add about your experience with the tasks or the lighting conditions?

B.1.7 Trigger words. We defined trigger words to look out for:

Rushed, annoyed, distracted, surprised, supported, flow, comfortable, easier/harder, overstimulated, faster/slower (time related terms), prettier/... (aesthetic terms)

B.2 Transcription Data

The interview transcripts⁵ and raw data for the thematic analysis⁶ are available on Github.

C Design Documents

The study design documents underlying this report are available on Github⁷.

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⁵<https://github.com/johannesgaebler/tuw-urm-report/blob/main/Data/InterviewTranscripts.txt>, accessed on 14.01.2026

⁶<https://github.com/johannesgaebler/tuw-urm-report/blob/main/Data/ThematicAnalysis.xlsx>, accessed on 17.01.2026

⁷<https://github.com/johannesgaebler/tuw-urm-report/tree/main/DesignDocuments>, accessed on 18.01.2026